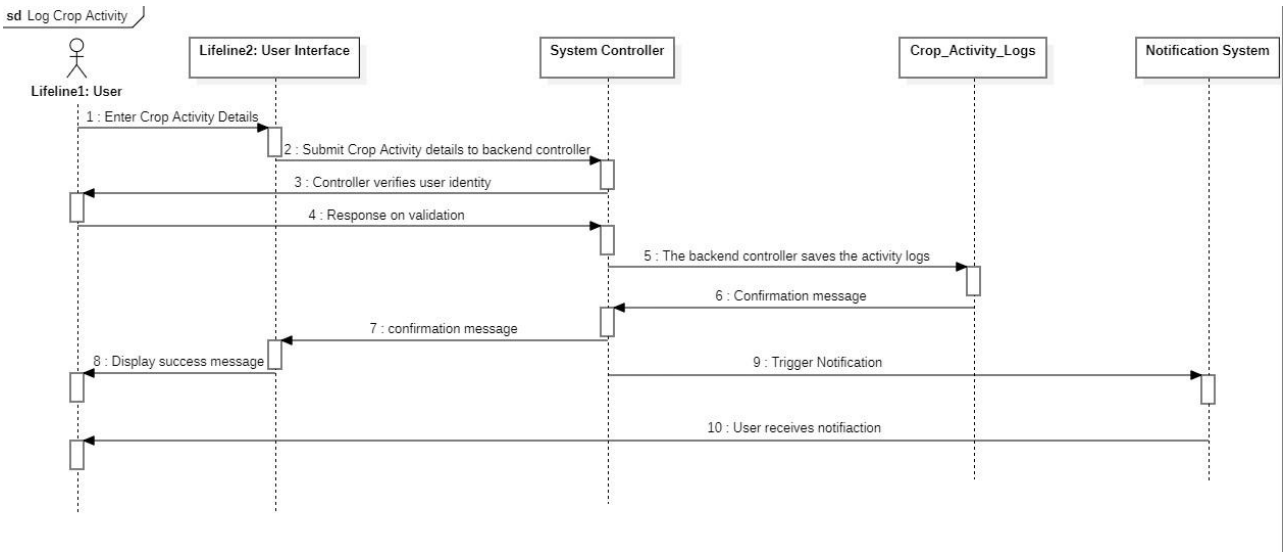
**SOFTWARE LAB-04**

**Name:** Roshini Pininti

**Register Number**

23BCE5129 **Sequence**

**Diagram**




This sequence diagram represents the step-by-step process for logging crop activity within a system, showing the interaction between the User, User Interface (UI), System Controller, Crop Activity Logs, and Notification System.

**Participants:**

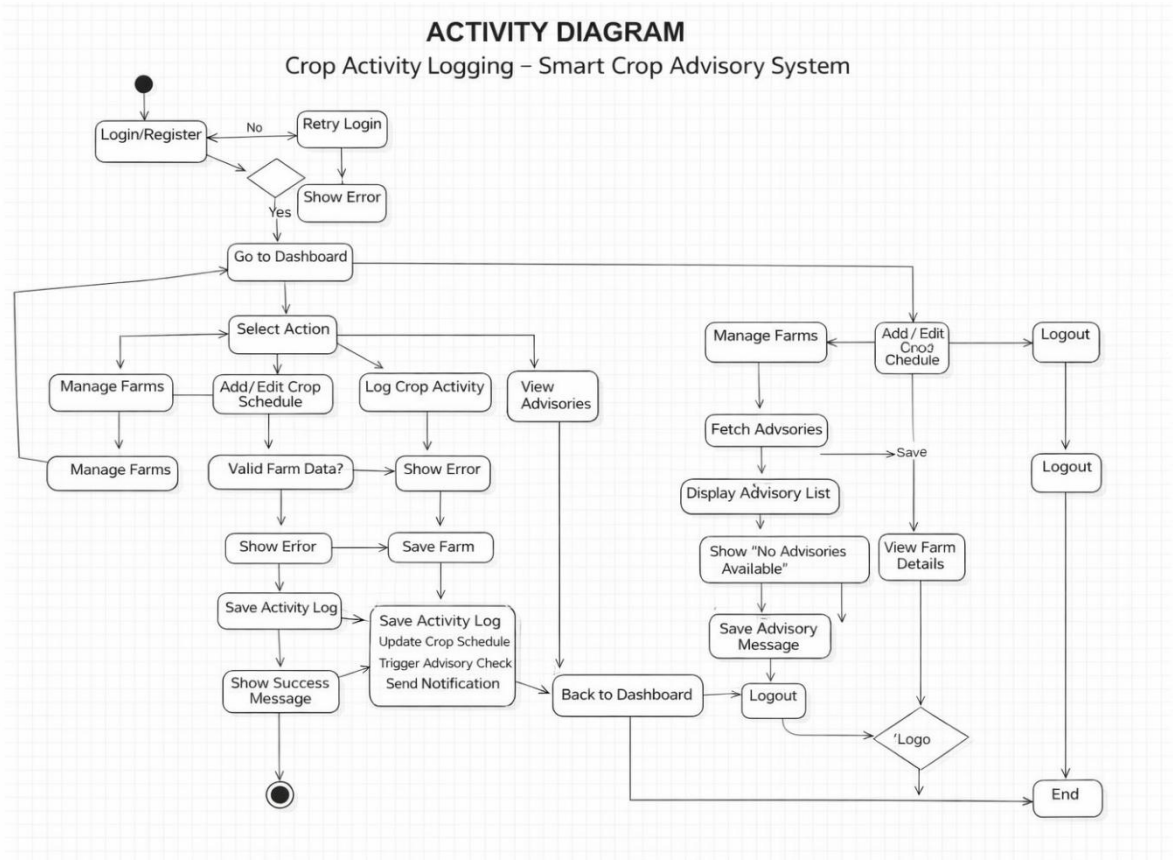
- **User:** The individual initiating the crop activity logging.
- **User Interface (UI):** The front-end system interface through which the user interacts.
- **System Controller:** Backend component that manages validation, data processing, and coordination.
- **Crop Activity Logs:** The data storage system where crop activity details are saved.
- **Notification System:** Responsible for sending notifications to the user.

**Flow of Activities:**

1. User Input: The process starts with the User entering crop activity details through the User Interface (Step 1: Enter Crop Activity Details).
2. Submission to Backend: The User Interface sends the entered crop activity details to the System Controller for processing (Step 2: Submit Crop Activity details to backend controller).
3. User Validation: The System Controller verifies the identity of the user to ensure they are authorized to perform this action (Step 3: Controller verifies user identity).
4. Validation Response: The validation result is sent back from the System Controller to

5. Display Success: The User Interface displays a success message to the User confirming that the crop activity has been recorded (Step 8: Display success message).
6. Trigger Notification: The System Controller triggers a notification to the Notification System about the newly logged crop activity (Step 9: Trigger Notification).
7. User Notification: Finally, the Notification System sends a notification to the User, completing the process (Step 10: User receives notification).
8. the User Interface (Step 4: Response on validation).
9. Saving Activity Logs: Once the user is validated, the System Controller saves the crop activity logs to the Crop Activity Logs system (Step 5: The backend controller saves the activity logs).
10. Confirmation Message: The Crop Activity Logs sends a confirmation message back to the System Controller indicating successful saving of data (Step 6: Confirmation message).
11. Notify UI: The System Controller sends a confirmation message to the User Interface to inform that the crop activity has been successfully logged (Step 7: Confirmationmessage).

## Activity Diagram




- The activity diagram represents the workflow of the Crop Activity Logging process in the Smart Crop Advisory System.

- The process starts with Register/Login**.**
- If credentials are invalid, the system displays an error message and asks the user to retry.
- If credentials are valid, the user is redirected to the Dashboard**.**
- From the dashboard, the user can select actions such as:
    - Manage Farms
    - Add/Edit Crop Schedule
    - Log Crop Activity
    - View Advisories
    - Logout
- When Log Crop Activity is selected, the user enters farm details, crop schedule, activity type, description, and date.
- The system validates the entered data.
- If the input is invalid, an error message is displayed and the user retries.
- If the input is valid, the system saves the activity log and updates the crop schedule.
- The system may trigger advisory checks and send notifications if required.
- After successful completion, a confirmation message is displayed, and the user is redirected back to the dashboard.
- The process ends when the user selects Logout**.**